

MO I RANA, Norway

WMO: 011510

Lat: 66.364N

Lon: 14.301E

Elev: 70

StdP: 100.49

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.9	-19.8	-25.4	0.4	-22.0	-22.9	0.5	-19.1	12.0	-1.5	10.2	-2.0	1.1	330	0.676

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	25.6	16.6	23.0	15.4	20.9	14.3	17.4	23.9	16.0	21.6	14.8	19.7	3.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.8	10.6	19.1	13.5	9.7	17.6	12.5	9.1	16.6	49.2	24.2	45.0	21.6	41.7	19.8	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.4	6.5	DB	-25.2	28.7	2.5	3.1	-27.0	30.9	-28.4	32.7	-29.8	34.5	-31.6	36.8	(4)
(5)			WB	-25.7	18.9	2.5	2.1	-27.5	20.4	-28.9	21.6	-30.3	22.8	-32.1	24.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-6.3	-5.6	-3.3	1.4	6.8	10.6	14.6	13.1	9.5	3.4	-1.5	-3.4	(6)	
	DBStd	8.44	6.71	6.50	4.89	3.34	3.79	3.18	3.74	2.79	3.20	3.61	4.78	5.52	(7)	
	HDD10.0	2776	505	436	412	258	117	29	1	4	47	206	344	417	(8)	
	HDD18.3	5493	763	669	670	508	358	233	129	165	264	463	594	675	(9)	
	CDD10.0	339	0	0	0	0	18	47	142	99	32	1	0	0	(10)	
	CDD18.3	15	0	0	0	0	0	1	13	2	0	0	0	0	(11)	
	CDH23.3	191	0	0	0	0	8	6	157	21	0	0	0	0	(12)	
	CDH26.7	36	0	0	0	0	1	0	35	1	0	0	0	0	(13)	
(14) Wind	WSAvg	2.5	2.7	2.7	2.7	2.5	2.9	2.8	2.4	2.3	2.3	2.4	2.3	2.4	(14)	
(15) Precipitation	PrecAvg	1461	156	144	144	86	69	77	86	92	144	162	131	171	(15)	
	PrecMax	1922	380	302	287	215	114	155	204	176	445	369	262	391	(16)	
	PrecMin	998	5	9	36	18	29	13	36	24	32	35	9	23	(17)	
	PrecStd	227	100	78	77	50	22	43	39	42	89	75	81	102	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.5	5.7	6.8	15.3	23.6	25.0	30.0	25.8	20.7	14.1	9.2	6.3	(19)	
		MCWB	5.2	4.3	3.6	8.2	13.4	15.6	19.2	16.4	14.6	10.0	6.8	4.7	(20)	
	2%	DB	5.0	4.6	5.2	10.3	20.5	21.7	27.0	23.4	18.1	12.1	7.7	5.1	(21)	
		MCWB	3.6	2.9	3.0	5.0	12.0	13.9	17.3	15.8	12.7	8.3	6.0	2.6	(22)	
	5%	DB	3.6	3.3	4.1	8.0	17.5	19.6	24.7	21.1	16.4	9.9	6.2	3.9	(23)	
		MCWB	2.0	1.7	2.1	4.1	10.8	12.6	16.7	15.1	12.0	7.6	4.3	1.8	(24)	
	10%	DB	2.0	2.1	2.9	6.4	13.8	17.5	22.1	19.1	14.8	8.2	4.6	2.7	(25)	
		MCWB	0.6	0.6	1.0	3.1	8.4	11.6	15.2	14.0	11.2	6.5	3.0	1.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	4.8	4.8	8.2	13.9	16.1	20.6	17.9	15.2	10.5	7.6	5.5	(27)	
		MCDB	6.3	5.5	5.7	14.3	22.8	22.3	27.6	22.9	18.7	13.7	8.5	5.9	(28)	
	2%	WB	3.8	3.3	3.6	6.1	12.4	14.4	18.5	16.3	13.6	8.9	6.1	3.1	(29)	
		MCDB	4.4	4.2	4.9	8.4	19.7	20.6	25.9	22.0	17.0	11.0	7.3	4.2	(30)	
	5%	WB	2.0	1.9	2.3	4.7	10.8	13.2	16.9	15.2	12.6	7.8	4.5	2.0	(31)	
		MCDB	3.3	2.9	3.6	7.0	17.4	18.5	23.3	20.2	15.6	9.3	6.0	3.3	(32)	
	10%	WB	0.7	0.8	1.2	3.6	8.9	12.0	15.7	14.3	11.5	6.8	3.0	1.2	(33)	
		MCDB	1.5	1.8	2.5	5.8	12.9	16.4	21.1	18.4	14.3	8.1	4.4	2.4	(34)	
(35) Mean Daily Temperature Range	MDBR		6.3	6.6	7.9	7.8	8.6	8.2	8.7	8.7	7.8	5.5	4.6	5.8	(35)	
	5% DB	MCDBR	5.5	5.5	6.8	11.0	14.9	13.0	14.3	13.2	10.4	7.0	5.5	6.0	(36)	
		MCWBR	4.8	4.7	5.2	7.0	7.8	6.7	6.3	6.7	6.1	4.9	4.3	5.0	(37)	
	5% WB	MCDBR	6.0	5.3	6.0	8.7	13.9	11.4	12.5	12.0	9.2	5.7	5.1	5.4	(38)	
		MCWBR	5.5	4.8	4.8	6.0	7.4	6.1	5.9	6.4	5.8	4.4	4.3	4.5	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.203	0.270	0.307	0.335	0.342	0.342	0.353	0.339	0.316	0.285	N/A	0.209	(40)	
	taud		1.920	2.103	2.134	2.169	2.263	2.381	2.424	2.465	2.506	2.284	N/A	1.933	(41)	
	Ebn at Noon		203	568	736	817	854	866	843	819	756	560	N/A	16	(42)	
	Edh at Noon		17	54	88	110	112	102	96	83	63	45	N/A	3	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.07	0.45	1.74	3.58	4.79	4.94	4.49	3.39	1.80	0.65	0.14	0.01	(44)	
	RadStd		0.01	0.09	0.25	0.31	0.29	0.48	0.53	0.36	0.21	0.11	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (25 neighbors)		+0.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page